



UNIVERSITY OF THE PUNJAB

Seventh Semester - 2015

Examination: B.S. 4 Years Programme

Roll No. 118

PAPER: Research Methodology
Course Code: STN15406

TIME ALLOWED: 2 Hrs
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Write short answer on the following questions:

1. Differentiate between bibliography and references.

2. Sampling frame and its effect on survey.

3. Write a note on cluster sampling.

4. Define pretesting of a questionnaire. What are the advantages?

5. Explain the term "Survey" and its various types.

6. What are the qualities of "good data"? Explain its types and how to collect data.

7. How sample size is determined under different conditions?

i) If the interval is narrow, sample size must be large.

ii) A large number of subgroups of stratified data.

iii) The greater sample must be taken.

UNIVERSITY OF THE PUNJAB

Seventh Semester 2016

Examination: B.S. 4 Years Programme

Roll No. 7454

Research Methodology
Code: STAT-406

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Write short answer on the following questions

(20)

- a) Nominal and Ordinal scale (82)
- b) Semantic differential and Likert scale (93)
- c) Cross sectional and longitudinal study
- d) Close ended questions

2. Define and explain under what kind of condition they are used.

(15)

- a) Stratified Sampling (134)
- b) Systematic Sampling (137)
- c) Simple random sampling (134)

3. Explain the errors that affect accuracy. Also, give the reasons how they can be avoided.

(15)

PUACP



UNIVERSITY OF THE PUNJAB

Seventh Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Research Methodology
Course Code: STAT-406

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

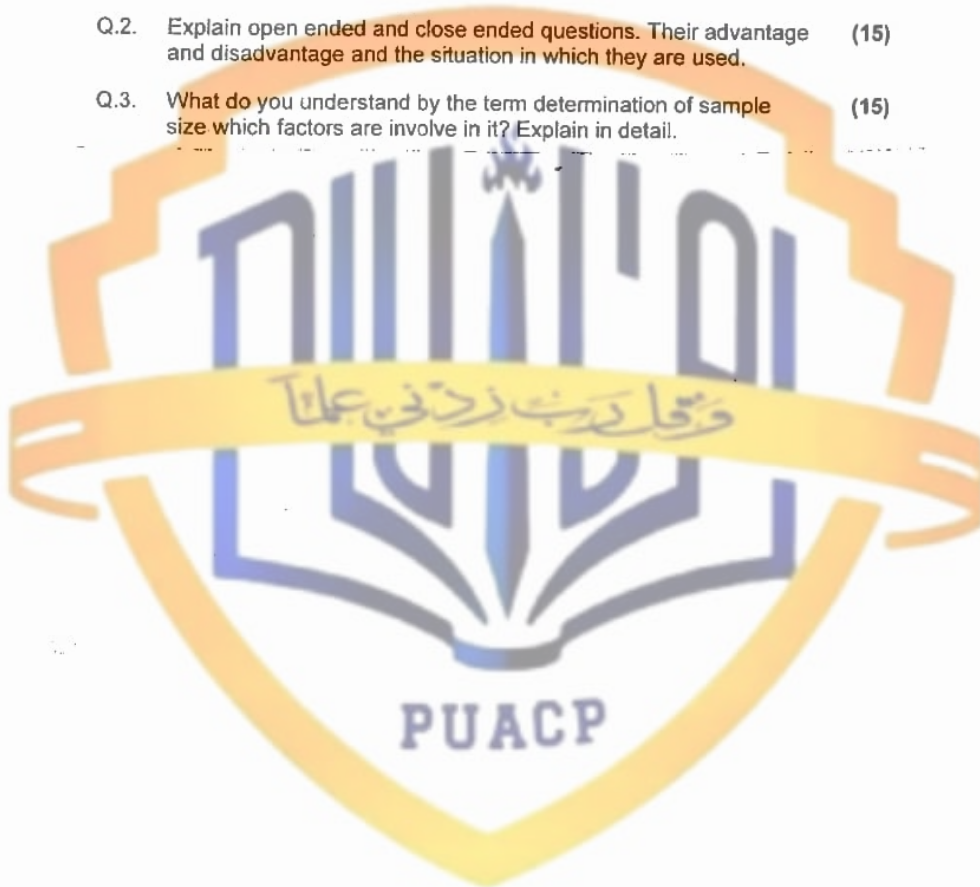
Subjective

Q1. Write short answer on the following questions. (20)

- a) Sampling frame
- b) Census
- c) Pretesting of questionnaire
- d) Literature review

Q2. Explain open ended and close ended questions. Their advantage and disadvantage and the situation in which they are used. (15)

Q3. What do you understand by the term determination of sample size which factors are involve in it? Explain in detail. (15)





UNIVERSITY OF THE PUNJAB

Seventh Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Research Methodology
Course Code: STAT-406

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

- Q1. Write short notes on (20)
- a) Stratified sampling
 - b) Measurement error
 - c) Processing of survey data
 - d) Types of data collection
- Q2. What is a questionnaire? Which factors should be considered in designing of a questionnaire? Explain in detail. (15)
- Q3. Define Scales and its types. Also explain different types of rating scales by giving at least two examples each. (15)





UNIVERSITY OF THE PUNJAB

Seventh Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Research Methodology

Course Code: STAT-406 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Write short answers on the following questions.

(4x5=20)

- a) Likert scale and Itemized Rating Scale
- b) Systematic Sampling
- c) Coverage and Measurement Error
- d) accuracy in survey

Q.3. Explain ten steps for a successful survey.

(10)

Q.4. Write in detail about the essential components of good research proposal and what are the factors affecting the research design?

(10)

Q.5. What are various types of questions asked while conducting a research? Discuss their advantages and disadvantages?

(10)



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – Fall 2021

Paper: Research Methodology

Course Code: STAT-406

Roll No.

Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions:

(6x5=30)

- (i) Sampling Frame and its requirements
- (ii) Criteria of good Research
- (iii) Test of validity for a measurement
- (iv) Difference between Questionnaire and Schedule
- (v) Significance of Report Writing
- (vi) Open Ended Questions

Answer the following questions.

(3x10=30)

- Q.2 Explain the meaning of research also describe the various types of research.
- Q.3 What do you mean by Research Design? What point should be taken into consideration by a researcher in developing a sample design for a research project?
- Q.4 What are the techniques for interpretation of research results? Describe the precautions that researcher should take while interpreting his findings.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2022

Paper: Research Methodology

Course Code: STAT-406

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- (i) Explain the sources of secondary data.
- (ii) What do you mean by telephonic surveys? Discuss their advantages and disadvantages.
- (iii) Explain discriminant analysis.
- (iv) Differentiate between schedule and questionnaire.
- (v) Discuss the different types of rating scale.
- (vi) What is difference between references and bibliography?

Answer the following questions.

(3x10=30)

- Q.2 Explain the types of error in sample survey. Describe the different methods to control each type of error.
- Q.3 What is literature review? Write its importance and functions. Explain the procedures for reviewing the literature.
- Q.4 What is pretesting of questionnaire? Write the advantages and disadvantages of pretesting of questionnaire.

PUACP



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2022

Paper: Time Series Analysis-I

Course Code: STAT-411

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

- Discuss the duality between moving average and auto-regressive process.
- Write a short note on transformation.
- Define ARIMA process.
- Discuss the pattern of ACF and PACF of MA(q) process with the help of graphs.
- Define secular trend and cyclical variation.
- Define correlation and partial Autocorrelation.

Answer the following questions. (3x10=30)

Q.2. Determine the trend line by the least square method from the following data. Plot the actual values and linear trend on the same graph.

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023
Price in Rs.	3	6	2	10	7	9	14	12	18

Q.3. Prove that for a stationary time series $\text{Corr}(X_t, X_{t-1}) = \text{Corr}(X_t, X_{t+1})$

Q.4. Two models of MA(1) process are stated below:

$$\text{Model-1 : } X_t = Z_t + \beta Z_{t-1}$$

$$\text{Model-2 : } X_t = Z_t + \frac{1}{\beta} Z_{t-1}$$

Where $\{Z_t\}$ is a purely random process with zero mean and finite variance σ^2

Find the autocorrelation functions of Model-1 and Model-2. Also comment on the invertibility condition of MA process by discussing the autocorrelation functions of both models.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Roll No. [REDACTED]

Paper: Research Methodology

Course Code: STAT-406

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- (i) Write advantages and disadvantages of open ended questions.
- (ii) Explain the meaning of research. Write the main objectives of research.
- (iii) What is test of validity? Differentiate between content validity and criterion-related validity.
- (iv) What is descriptive research? Write its goals.
- (v) Explain non-response error and its sources. What methods would you suggest to control such type of error?
- (vi) What is pretesting of a questionnaire? What are the aims of the pretesting of a questionnaire?

Answer the following questions.

(3x10=30)

- Q.2** What is the meaning of measurement in research? What difference does it make whether we measure in terms of a nominal, ordinal, interval or ratio scale? Explain by giving examples.
- Q.3** What is review of literature? Explain the objectives, need and functions of review of literature.
- Q.4** What is the importance of research report? Describe, in brief, the major sections involved in writing a research report.



UNIVERSITY OF THE PUNJAB
B.S. 4 Years Program / Seventh Semester – Fall 2023

Paper: Research Methodology
Course Code: STAT-406

(Common with BS-AD 7th)

Roll No. 11UG133

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- (i) Write the advantages and disadvantages of mail surveys.
- (ii) Write a short account on stratified random sampling.
- (iii) Describe the precautions that researcher should take while interpreting his findings.
- (iv) Explain bibliography and write its importance in context of research report.
- (v) What are the characteristics of research process?
- (vi) Explain quantitative research and qualitative research with the help of examples.

Answer the following questions.

(3x10=30)

- Q.2 What do you mean by research design? Differentiate between research methodology and research design. Illustrate your answer with suitable examples.
- Q.3 Discuss the qualities of well-designed questionnaire. Describe the various steps which are used in designing a questionnaire.
- Q.4 Explain the essential components of good research proposal? What are the major sections of writing a research proposal?